PAYTM BUILD FOR INDIA AI HACKATHON · BENGALURU FINALS · 31 OCT 2026

Saathi

An autonomous service and sales teammate for Paytm merchants.

Resolves the problem before a ticket exists.
Closes a first-time loan with a human on the money step.

Track 3: Autonomous AI Teammates

Team Saathi · Amit Mishra · built solo

Partner stack: Sarvam AI (saarika, bulbul, sarvam-105b) live in the demo

SAATHI'S SCOREBOARD

One merchant, one morning, before he answered the phone

₹1,340 Recovered
Soundbox restarted, 7 payments confirmed

₹2,300 Collected
Udhaar reminder on WhatsApp, paid

₹50,000 Offered
Diwali loan, approved by a human

Demo data from the 30 Oct rehearsal; merchant backend simulated

PROBLEM

Tuesday, 6:10 PM, Rajajinagar

6:10 PM The Soundbox stops announcing. The light blinks green. Nobody in the shop notices.

6:50 PM ₹2,340 has landed in the app, unannounced. Forty minutes of silence at the busiest hour.

Lost Two customers walked out, sure their payment had failed. The owner finds out tomorrow, if at all.

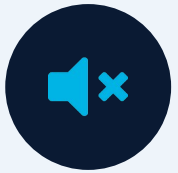
Paytm's own support pages say so

- The Soundbox announces successful payments only, never failed or pending ones.
- Settlement SMS: “due to network issues you might not receive” it. Wait two days, then phone the helpline.
- Offline device: read the LED colour, reseat the SIM, call 0120-4440440.

business.paytm.com/support · paytm.com/blog/paytm-help

Support waits for the ticket. The merchant's money has no owner.

Three leaks. Nobody owns them.



Silent device

Payments land but nobody hears them. Customers walk, or pay twice.



Stuck settlement

Money sits in Pending or Failed past T+1. The merchant waits, then queues at the helpline.



Ageing udhaar

Credit given on trust, written in a notebook, chased in person. The oldest lines are never collected.

SOLUTION

Triggered by the event, not by a question

Saathi wakes on the merchant's problem, works it in the merchant's language, and escalates to a human whenever money moves.



Watch

Device silent, settlement stuck, udhaar ageing

Diagnose

Root cause from the data, in rupees at stake

Act

Restart, retry, remind the customer, offer credit

Confirm

Calls the merchant in Hindi or Kannada; tells, asks, listens

Learn

Every outcome logged with who did it and why it was allowed

Reaches the merchant's customers

For udhaar, Saathi messages the customer on WhatsApp in their language with the merchant's verified P2M UPI link. Never a P2P collect request (withdrawn 1 Oct 2025). The customer pays; the ledger closes itself.

Credit only through the gate

A loan is mentioned only when the ledger shows a cash gap and the merchant has consented. Saathi reads the lender, APR and Key Fact Statement aloud, logs consent, keeps the cooling-off, and raises a human approval. It cannot disburse.

TRAI

AI disclosed in the first sentence. Calls only 9 AM to 7 PM. No loan pitch on a service call.

RBI Digital Lending 2025

Lender named, KFS read, cooling-off kept, no automatic limit increase.

DPDP

Consent per purpose, timestamped, revocable, shown on the dashboard.

FREE-AI

Every rupee spoken is a deterministic sum; the log shows the line behind it.

The demo: Ravi's phone rings. Soundbox fixed before he answers, ₹2,300 udhaar collected on WhatsApp, a ₹50,000 loan approved by a human.

TECH STACK

Sarvam AI is the voice; code holds the rules

REAL SIMULATED BUILT, NOT LIVE

Merchant data

SIMULATED

- 20 seeded merchants
- Soundbox events
- Settlements
- Udhaar ledger
- Loan eligibility
- Swap to Paytm APIs by config

Saathi agent on Sarvam AI

REAL

Listen saarika:v2.5
speech-to-text, hi-IN and kn-IN

Think sarvam-105b
tool-calling LLM, decides every step

Speak bulbul:v3
text-to-speech, streamed by sentence

Tool layer, typed registry. One role per tool: Diagnoser, Fixer, Seller, Compliance, Manager. A compliance ticket gates send_reminder and offer_loan in code, whatever the model says.

REAL

Human approval card on the dashboard; mandatory above ₹25,000. The agent has no disburse tool.

Channels

REAL
Browser call screen (plan A)

BUILT, NOT LIVE
Twilio Media Streams; Exotel next

REAL
WhatsApp utility message (Twilio)

REAL
Live ₹ dashboard (server-sent events)

Measured 30 Oct

1.5 to 3.2 s from end of speech to first word back. The opening briefing is prepared while the phone rings.

Tested

Typecheck and 86 unit tests in 12 files. Live LLM tests in Hindi and Kannada assert compliance runs before every gated action.

Swappable

Node and TypeScript. Every tool has an HTTP surface, so a real Paytm API drops in without touching the agent.

USP

The triple nobody else ships

Event-triggered outreach · reaches the merchant's customers · consent-gated credit

	Proactive, event-triggered	Reaches the merchant's customers	Acts, not just answers	Consent-gated credit
Saathi	Yes. Wakes on device silent, settlement stuck, udhaar ageing	Yes. WhatsApp with the merchant's P2M UPI link	Yes. Restarts, retries, reminds, offers; human on the money step	Yes. Consent, KFS read, cooling-off, human approval
BharatPe Agentic AI Sep 2026, in-app	Partly. Follows up on requests the merchant raised	No	Yes. Chargebacks, 60+ systems	Partly. Pushes Credit-on-UPI offers in-app
Paytm AI Soundbox Oct 2025, 11 languages	No. Answers questions on the device	No	No. Reports payments and performance	No
Razorpay RAY Sep 2026, WhatsApp	Partly. Issues alerts, online PG only	No	Partly. Checks settlements, answers	No
Khatabook IVR paid robocalls, SMS	No. Merchant schedules the blast	Yes. A recorded call, no conversation	No. Same message to everyone	Partly. In-app NBFC loans, not agent-led

Compliance is the tiebreaker, and it is on screen, not on a slide. TRAI: AI disclosed, 9 AM to 7 PM, no loan pitch on a service call. RBI Digital Lending Directions 2025: KFS, cooling-off, no auto limit increase. DPDP: consent per purpose. FREE-AI: every number explainable. The dashboard shows consent time, KFS read, APR, cooling-off, and a greyed-out Disburse the agent cannot press.

Other voice bots wait for questions. Saathi hunts for lost money and acts.

IMPACT AND BENEFITS

Measured in Paytm's own numbers

1.64 crore

storefronts with a Soundbox. Saathi works over phone and WhatsApp, so every one of them, not only the new AI device.

Paytm Q2 FY27

~7%

of subscription merchants have taken a loan; over half of disbursements go to repeat borrowers. The first-time borrower is the gap.

Paytm Q4 FY26 call, Q2 FY27

₹912 crore

financial-services distribution revenue in one quarter, up 49% YoY. Saathi feeds this funnel at the moment of need.

Paytm Q2 FY27

Top 2

complaint clusters in Paytm's own support pages and store reviews: Soundbox silence and settlements. Saathi's first two triggers.

business.paytm.com, Play Store

Pilot targets (targets, not measurements)

- **Tickets deflected** Soundbox and settlement issues resolved before the merchant calls
- **Udhaar recovered** Rupees collected per store per month, attributed to a Saathi reminder
- **First-time borrowers** Consented, KFS-read, human-approved offers against the 7% base

One merchant, one month: a worked example

Takings ₹14,200 a day (demo seed), about ₹1,000 per trading hour.

A1 (assumption): Soundbox silent one hour a week; one customer in four walks out. ₹250 a week, ₹1,000 a month protected.

A2 (assumption): Unrecovered udhaar about ₹8,200 a month per store (vendor survey, estimate); reminders recover a quarter. ₹2,050 a month collected.

Cost: 10 short calls at ₹3 to 5 plus 30 WhatsApp messages at ₹0.115 (list prices, estimate). Under ₹50 a month.

About ₹3,000 kept for about ₹50 spent. Loan income excluded: the per-loan take rate is not public.

BUSINESS MODEL

A ₹30 to 50 add-on that pays for itself

Merchant add-on

- ₹30 to 50 a month on top of the Soundbox subscription (merchants already pay about ₹100 a month for the audio device).
- Or inference-fee pass-through, the model Paytm has announced for the AI Soundbox.
- Voice for exceptions only; a WhatsApp utility message for the daily summary.

Unit economics

- A three-minute call: ₹3 to 5 at Sarvam list prices (STT ₹0.49/min, TTS ₹1 per 1k chars, telephony ₹0.7 to 1.15/min). Estimate.
- A WhatsApp utility message: ₹0.115.
- Ten calls and thirty messages a month fit inside the fee with margin to spare.

Lending revenue share

- Saathi is a distribution surface, never the lender. Loans come from Paytm's existing 14 lending partners.
- Paytm keeps its distribution revenue (₹912 crore a quarter today) on every first-time borrower Saathi converts.
- No limit changes, no disbursal, no collections by the agent.

ROLLOUT

Today → **Pilot: one city** → **Scale**

Live Hindi and Kannada agent on Sarvam, three triggers, simulated merchant backend, WhatsApp sandbox.

Shadow mode with human review on every action. Scored in rupees recovered, collected and offered.

All 1.64 crore Soundbox storefronts over phone and WhatsApp, 10+ languages, Exotel as carrier.

BACKUP

What the LLM is not allowed to do

The model chooses words and the next tool. Everything else is deterministic code, and the demo shows it.

Never

- No arithmetic. Every rupee it speaks is a sum computed by the backend and passed in as text.
- No money movement. There is no disburse, refund or transfer tool in the registry.
- No credit limit changes, ever; no automatic top-ups.
- No loan mention on a service call unless the ledger shows a cash gap and the merchant has consented.
- No action outside the typed tool registry; a tool call the compliance ticket has not cleared is refused in code.
- No call outside calling hours and no call to a DND number; the tool layer refuses regardless of the prompt.

Deterministic triggers and gates

Device silent: no announcement for more than 30 minutes while the app shows inflows

Settlement stuck: status Pending or Failed past T+1

Udhaar ageing: a ledger line older than 10 days (demo: 12 and 15 days)

Reminder gate: 6 checks before a send, including consent, calling hours, DND and the approved template

Loan gate: 9 checks before an offer, including eligibility, lender named, KFS, explicit consent and cooling-off

Human approval: mandatory above ₹25,000; the Approve button is on the dashboard, not in the agent